

1. Leader Election Timing

In the Raft consensus protocol, follower election timeouts are randomized between 150 ms and 300 ms to prevent split-vote scenarios. If a follower receives no heartbeat AppendEntries RPC within this window, it transitions to Candidate state and increments the current term.

2. Heartbeat Frequency

The cluster leader broadcasts empty AppendEntries RPC heartbeats every 50 ms to maintain leadership authority and prevent followers from timing out.

3. Quorum Requirements and Split-Brain Mitigation

For a cluster of N nodes, consensus requires a strict majority quorum formula of $(N/2) + 1$ nodes. In a 5-node cluster, at least 3 nodes must acknowledge a log entry before it is committed to state machine.

4. Log Compaction and Snapshots

When log entries exceed 10,000 records, the Raft engine triggers asynchronous state machine snapshotting, compacting preceding log entries and discarding applied index history.